FIITJEE

ALL INDIA TEST SERIES

OPEN TEST

JEE (Advanced)-2021

PAPER –2

TEST DATE: 24-01-2021

Time Allotted: 3 Hours **Maximum Marks: 198**

General Instructions:

- The test consists of total **54** questions.
- Each subject (PCM) has **18** questions.
- This question paper contains **Three Parts.**
- **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
- Each **Part** is further divided into **Three Sections**: **Section-A, Section – B & Section-C.**

Section-A (01 – 06, 19 – 24, 37– 42) *this section contains* ***18 multiple*** *choice questions.*
Each question has FOUR options. ***ONE OR MORE THAN ONE*** *of these four option(s) is (are) correct answer(s).*
For each question, choose the option(s) corresponding to (all) the correct answer(s)
Answer to each question will be evaluated according to the following marking scheme:

Full Marks	**: +4** If only (all) the correct option(s) is (are) chosen:
Partial Marks	**: +3** If all the four options are correct but ONLY three options are chosen;
Partial Marks	**: +2** If three or more options are correct but ONLY two options are chosen and both of which are correct;
Partial Marks	**: +1** If two or more options are correct but ONLY one option is chosen and it is a correct option;
Zero Marks	**: 0** If none of the options is chosen (i. e. the question is unanswered);
Negative Marks	**: –2** In all other cases

Section-B (07 – 12, 25 – 30, 43– 48) *contains* ***18 Numerical*** *based questions with* ***Single digit integer*** *as answer, ranging from* ***0 to 9*** *and each question carries* ***+3 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*

Section-C (13 – 18, 31 – 36, 49– 54) *contains* ***18 Numerical*** *answer type questions with answer XXXXX.XX and each question carries* ***+4 marks*** *for correct answer and* ***0 marks*** *for wrong answer.*

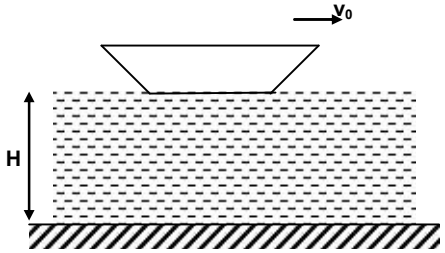
Physics

PART – I

SECTION – A
(One or More than one correct type)

This section contains **06** multiple choice questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

1. Water is flowing on a horizontal surface where velocity varies with height y from the ground as $v = \frac{v_0}{H^2}\left(2y^2 - \frac{y^3}{H}\right)$ where H is the depth of water. Coefficient of viscosity of water is η. Then choose the correct option(s).



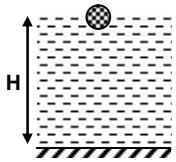
(A) Shear stress at $y = \frac{H}{2}$ is $\frac{5v_0\eta}{4H}$

(B) Shear stress at $y = H$ is $\frac{\eta v_0}{H}$

(C) Viscous force acting on the boat of base area A is $\frac{\eta A v_0}{H}$

(D) Viscous force acting on the boat of base area A is $\frac{\eta A v_0}{2H}$

2. A sphere of radius r and density ρ is released from the surface of a liquid of density $\frac{\rho}{2}$ and coefficient of viscosity η. It achieves terminal velocity just before reaching the ground. Then choose the correct option(s).



(A) Terminal velocity is $\frac{2r^2\rho g}{9\eta}$

(B) Terminal velocity is $\frac{r^2\rho g}{9\eta}$

(C) Magnitude of work done by viscous force on the sphere is $\frac{2\pi r^3\rho g}{3}\left(H - \frac{\rho^2 r^4 g}{81\eta^2}\right)$

(D) Magnitude of work done by viscous force on the sphere is $\frac{4}{3}\pi r^3\rho g\left(H - \frac{\rho^2 r^4 g}{9\eta^2}\right)$

3. The air column in a pipe closed at one end is made to vibrate in its first overtone by a tuning fork of frequency 550 Hz. The speed of sound in air is 330 m/s. End correction may be neglected. Let P_0 denotes the mean pressure at any point in the pipe and ΔP_0 the maximum amplitude of pressure variation in the pipe. Then choose the correct option(s).

(A) The length of the pipe is 45 cm.

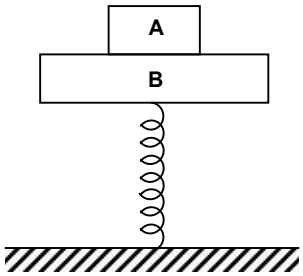
(B) The length of the pipe is 75 cm.

(C) The maximum pressure at the middle of the pipe is $\left(P_0 + \frac{\Delta P_0}{\sqrt{2}}\right)$

(D) The maximum pressure at the closed end of the pipe is $(P_0 + \Delta P_0)$

4. Coefficient of volume expansion of an ideal gas is $\gamma = \frac{2}{T}$ (T is the absolute temperature) in a process. C_P and C_V are the molar specific heat capacities at constant pressure and constant volume respectively. If the molar heat capacity of the process is C. Then choose the correct option(s).
(A) $C = C_V + 2R$
(B) $C = C_P + R$
(C) $C = 3C_V + 2R$
(D) $C = C_V - 2R$

5. Block A of mass m is placed over a plank B of mass 3 m. Plank is connected with a spring of force constant K. The system is oscillating. The block A just loses contact with the plank when plank is momentarily at rest. Then choose the correct option(s).



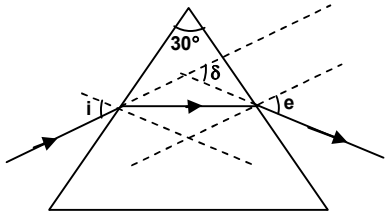
(A) Period of oscillation is $T = 2\pi\sqrt{\frac{m}{K}}$
(B) Period of oscillation is $T = 4\pi\sqrt{\frac{m}{K}}$
(C) Amplitude of oscillation is $A = \frac{4mg}{K}$
(D) Maximum compression in the spring is $\frac{8mg}{K}$

6. Light consisting of wavelength 400 nm to 800 nm falls on a metal plate A having work function 1.5 eV and saturation current 10mA. Now the same light is incident on a metal plate B having work function 2.0 eV. Take spectral radiancy of light is uniform and take hc = 1240 eV-nm. Then choose the correct option(s).
(A) Stopping potential for metal A is 3.1 volt
(B) Stopping potential for metal B is 1.1 volt
(C) Saturation current for metal B is 5.5 mA
(D) Stopping potential for metal A is 1.6 volt

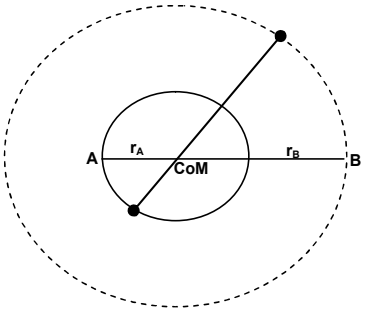
SECTION – B
(Single Digit Integer Type)

This section contains **06** questions. The answer to each question is a **Single Digit integer** ranging from **0** to **9**, both inclusive.

7. A ray of light is incident at an angle of incidence 60° on one face of a prism as shown in the figure. The angle of prism is 30°. The ray emerging out of the prism makes an angle of 30° with the incident ray. If refractive index of the prism is $\mu = \sqrt{a}$. Find the value of a.



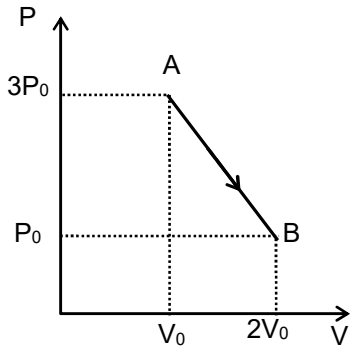
8. A binary star system consists of two stars - A (of mass 12 M_s) and B (of mass 2 M_s), where M_s is mass of the sun. They are separated by a distance d and they are rotating about their common centre of mass in circular orbits. Find the ratio of the kinetic energy of star B to that of the star A.



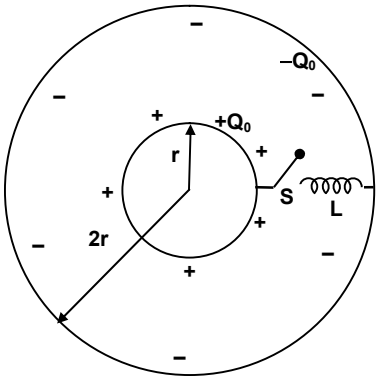
9. A conducting square loop of side 2.0 cm is rotated through 180° about one of its diagonals in 0.20 s. A uniform magnetic field exists in the region which is perpendicular to the plane of the loop initially. If average induced emf is 20 mV, find the magnitude of the magnetic field (in Tesla).

10. A thin uniform rod of mass 6 kg is pushed at lowest point along a rough horizontal plane by a constant horizontal force F = 80 N. The rod is in the pure translational motion and making an angle θ from the vertical. If angle θ in degree is 5k (coefficient of friction $\mu = \frac{1}{3}$, rod is in the vertical plane). Find the value of k.

11. One mole of an ideal monoatomic gas undergoes a process A $\rightarrow$ B as shown in the P-V diagram. If the process changes from endothermic to exothermic at the volume $k\left(\frac{5V_0}{16}\right)$ of the gas. Find the value of k.



12. Two concentric conducting spherical shells of radius r and 2r are having initial charges $+Q_0$ and $-Q_0$ respectively as shown in the figure. Switch S is closed at time t = 0. If current through the inductor L at time $t = \frac{\pi}{4}\sqrt{8\pi L\varepsilon_0 r}$ is $\frac{Q_0}{n\sqrt{\pi\varepsilon_0 Lr}}$. Find the value of n.

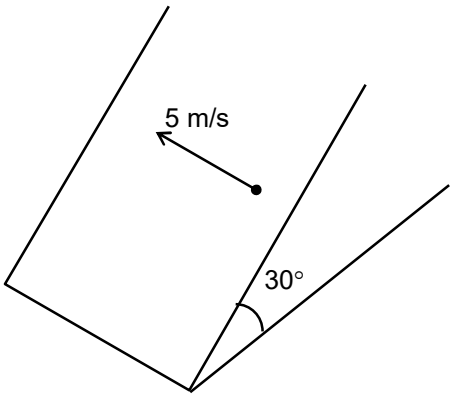


SECTION – C
(Numerical Answer Type)

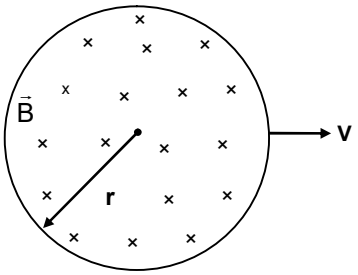
This section contains **06 questions**. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **SECOND DECIMAL PLACE**; e.g. XXXXX. XX).

13. A particle is projected from ground at an angle 37° with the horizontal. It collides at its highest point of trajectory with a very heavy vertical wall moving towards the particle with a constant horizontal velocity. The coefficient of restitution is $e = \frac{2}{3}$. Find the ratio of the particle's speed of projection to that of the wall so that after collision particle reaches the point of projection.

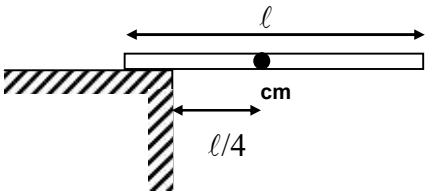
14. A particle is given a horizontal velocity of 5 m/s on a smooth inclined plane as shown in the figure. The plane has an angle of inclination 30° with the horizontal. Find the radius of curvature (in m) of its path after t = 1 s from starting. (Take $\sqrt{2} = 1.414$)



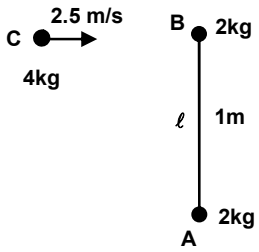
15. A uniform magnetic field exists in a circular region of radius R = 10 m. At time t = 0, a conducting circular loop of radius R = 10 m completely encloses the field. The circular loop starts moving in its plane and perpendicular to the magnetic field with a uniform velocity v = 5 m/s. Intensity of magnetic field is B = 20 mT. Find the induced emf (in volt) in the loop at time t = 2 s. (Take $\sqrt{3} = 1.73$)



16. A thin uniform rod is placed on a sticky table such that its mid-point is at one-fourth of its length from the edge of the table. It falls due to gravitational force and slips after rotating 45° from its initial horizontal position. Find the coefficient of friction between the rod and the table.



17. Two particles A and B each of mass 2 kg are connected by a massless rod of length ℓ =1m. Another particle C of mass 4 kg moving with a velocity 2.5 m/s (perpendicular to the rod AB) collides with the particle B as shown in the figure. Coefficient of restitution, e = $\frac{1}{2}$ and the system is placed in a gravity free space. Find the angular velocity (in rad/s) of the rod just after collision.



18. Parallel combination of an ideal choke coil and a resistance (R = 60Ω) is connected to an a.c source. The current through the choke coil is I_{Ch} = 3.0 A and through the resistance is I_R = 2.5 A. Find the net impedance (in Ω) of the circuit. (take $\sqrt{15.25} = 3.9$)

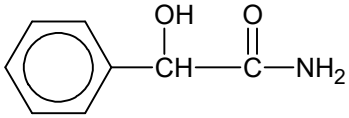
Chemistry

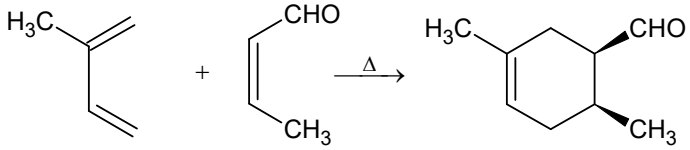
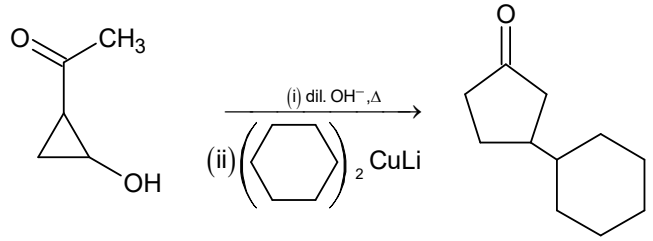
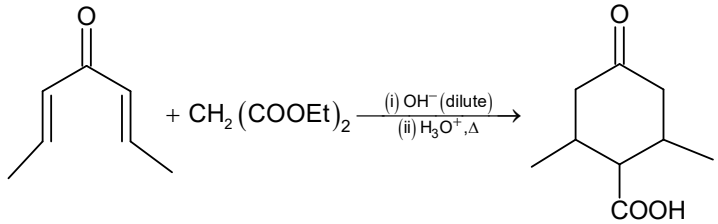
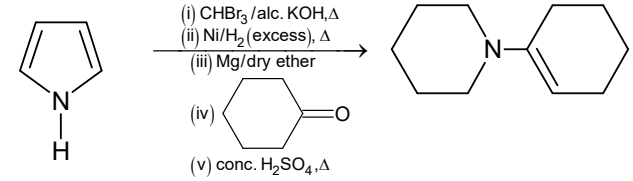
PART – II

SECTION – A
(One or More than one correct type)

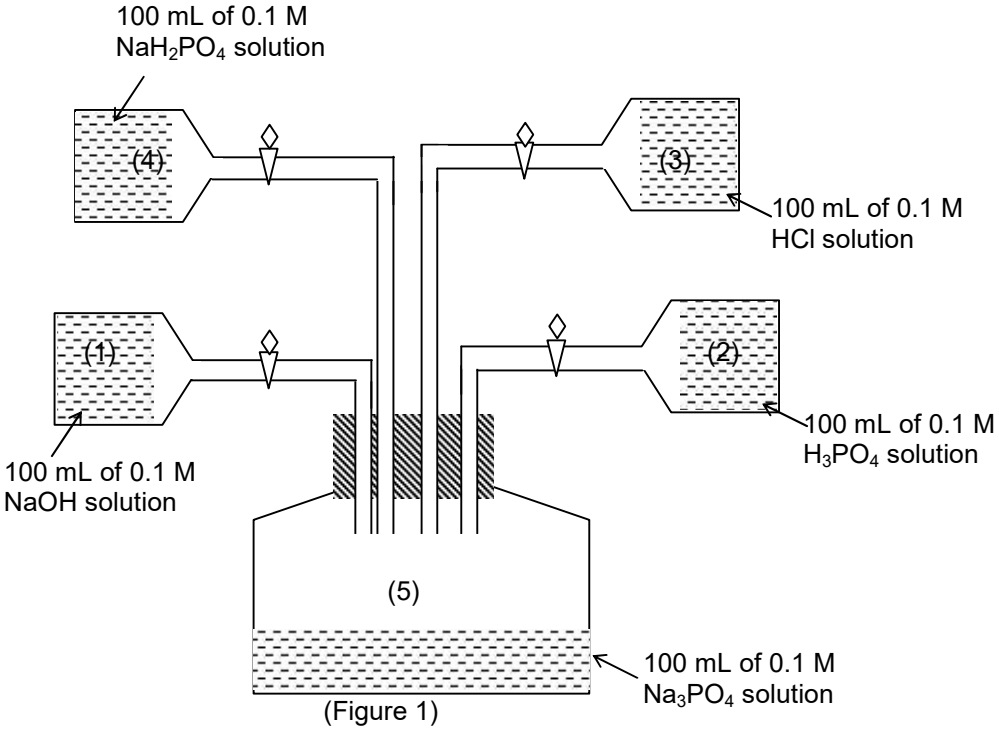
This section contains **06** multiple choice questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

19. Choose the correct option(s):
   (A) $K_2\left[Zn(Cl)(Br)(CN)(NO_2)\right]$ possesses 4 pair of enantiomers (consider all donor sites of ambidient ligands).
   (B) Ethylene diamine methyl ethylenediamineplatinum(II) chloride exists as a pair of enantiomer
   (C) $\left[Co(ox)(F)_2(Cl)_2\right]^{3-}$ posseses four stereoisomers (where 'OX' represents oxalate ion
   (D) $K_3\left[FeF_6\right]$ is a red-brown coloured complex

20. Identify the correct statement(s) of the following:
   (A) Potential energy of the half-chair conformer of cyclohexane is less than its twist boat conformer
   (B) Anti conformation of 2-fluoroethanol is less stable than its Gauche conformation
   (C)  on heating with NaBrO gives aniline as the major product
   (D) During the ozonolysis of 2-pentene, a total of 10 ozonides are formed (assume cross ozonides are also formed).

21. Which of the following reaction(s) is/are correctly matched with their major product?
   (A) 
   (B) 
   (C) 
   (D) 

22. Solutions from beaker number 1, 2, 3 and 4 are poured into beaker number (5) after opening the stop-cocks as instructed in the options (See Figure 1)



(Figure 1)

pK_{a_1}, pK_{a_2} and pK_{a_3} of H_3PO_4 are 3, 8 and 12 respectively.

Now, select the correct option(s):

(A) If stop-cock of only beaker no. (2) is opened then the pH of the solution in beaker (5) is 8

(B) If stop-cock of only beaker no. (2) and (4) are opened simultaneously, then the pH of the content in beaker (5) is 7.70

(C) If stop-cock of only beaker no. (2), (3) and (4) are opened simultaneously, then the pH of the content in beaker (5) is 5.5

(D) If the stop-cocks of all the four beakers (1), (2), (3) and (4) are opened simultaneously, then the pH of the content in beaker no. (5) will be 7.70
[While answering the question, assume the temperature is 298 K and the whole content of beakers are transferred into beaker (5)]

23. In the crystal of solid (x), A^{3+} ions are present at each of the corner and B^- ions are present at each of the face centre of a cubic unit cell. When 0.6 gm of the above solid is dissolved in 100 gm of water, solution showed freezing point of $-0.2976^\circ C$. The solute (x) is assumed to be 20% dissociated in aqueous solution and K_f of H_2O is $1.86\ K\ kg\ mol^{-1}$. Now select the correct option(s):

(Assume that $r_{A^{3+}} = 0.414\,\text{Å}$, $r_{B^-} = 1.0\,\text{Å}$ and assume element 'A' exists in +3 and +4 oxidation state and Avogadro's number $= 6\times10^{23}$)

(A) Molar mass of (x) is 60 g mol^{-1}

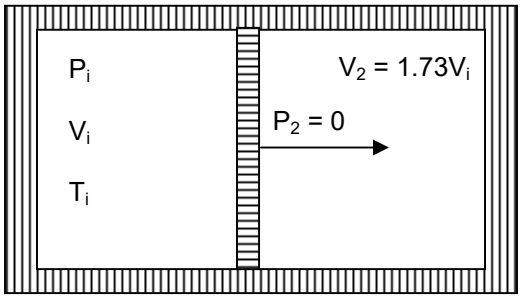
(B) Density of (x) is 1.25 g/cm^3

(C) Total number of unit cells of 'x' in solution are 6×10^{21}

(D) If the defected solid is found to contain 12% cation vacancies, then % of A^{4+} ions in its crystal is 40.90%

24. An ideal monoatomic gas is undergoing an irreversible change as shown in the figure:
(Assume the boundary to be thermally insulating and piston is massless, frictionless and moveable).

V_2 = Volume of second partition



P_i = Initial pressure

V_i = Volume of first partition

T_i = Initial temperature

P_2 = External pressure

T_2 = Final temperature

Now, select the correct option(s) regarding the above process: (Where T_2 = final temperature of gas)

(A) $T_2 = T_i$

(B) $P_1V_1^{\gamma} = P_2V_2^{\gamma}$

(C) $\Delta S_{syst} = \frac{P_iV_i}{T_i}$

(D) $\Delta S_{syst} = 0$

SECTION – B
(Single Digit Integer Type)

This section contains **06** questions. The answer to each question is a **Single Digit integer** ranging from **0** to **9**, both inclusive.

25. Consider the following species:

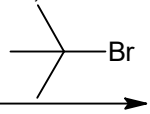
$\left[VO(acac)_2\right]$ (I) $\left[Fe(CO)_5\right]$ (II) $[PtCl_4]^{2-}$ (III) $K_4\left[Fe(CN)_6\right]$ (IV) $[IF_7]$ (V)

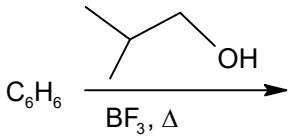
[Where acac : acetyl acetonate ion]
If, the sum of axial d-orbitals having conical nodal surfaces in all the species be 'x'
The sum of axial d-orbitals having two nodal planes in all species be 'y' and the sum of non-axial d-orbitals in all species be 'z' then find the value of $(x + z - y)^2$.

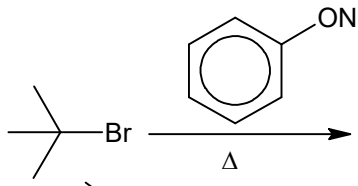
26. Total number of moles of ethanoyl chloride required to completely acetylate one mole of Lactose is…….

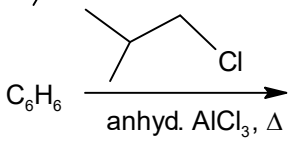
27. 1 L hard water (density = 1.5 g/cm^3) after passing through acidic resin (RH_2) has pH = 3.52 at 25°C. Assume that Ca^{2+} ions are the only cations present in hard water sample, if the degree of hardness of this sample of hard water in parts per million (ppm) of $CaCO_3$ is found to be 2x, then find the value of x. [$\log_{10}3 = 0.48$]

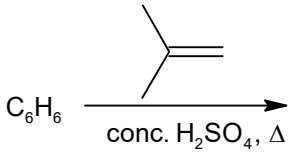
28. Some reactions are given below. The number of reaction(s) in which tert-butyl benzene is formed as the major product, is/are….

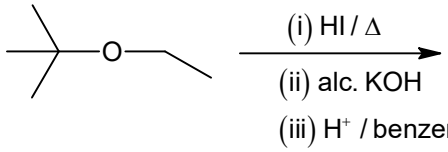
I. C_6H_6 +  $\xrightarrow[\Delta]{CH_3ONa}$

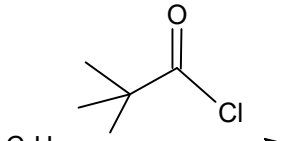
II. 

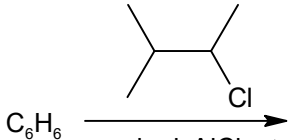
III. 

IV. 

V. 

VI. 

VII. 

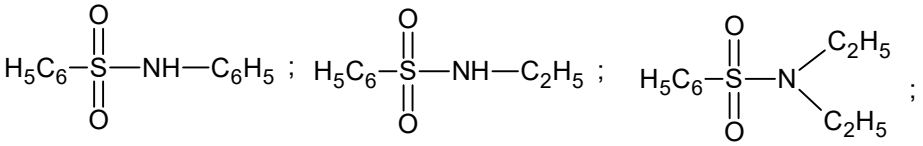
VIII. 

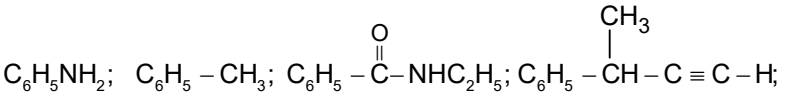
29. Among the following, total number of compound(s) which produce oxygen as the only gaseous product on gentle heating, is/are….

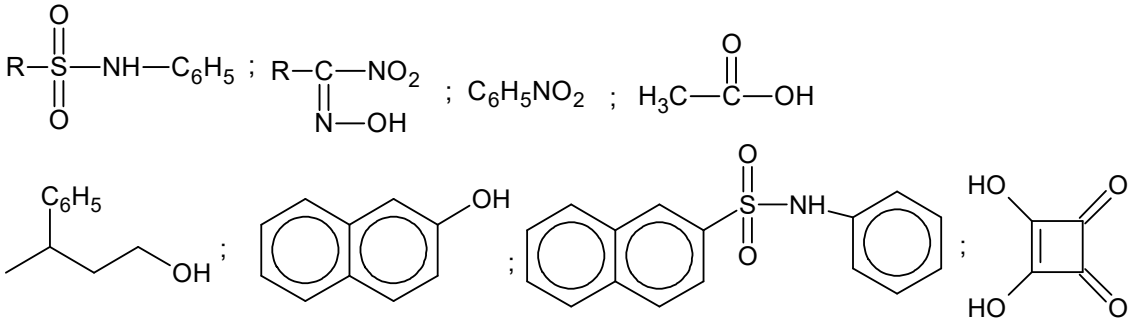
$NaNO_2$, KNO_3, Ag_2O, $Ca(NO_3)_2$, $KClO_3$, HgO,

CaO, Na_2O, $NaNO_3$, $Ba(NO_3)_2$, H_2O_2.

30. The number of compound (s) which is/are soluble in aqueous NaOH solution, is/are ….







SECTION – C
(Numerical Answer Type)

This section contains **06 questions**. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **SECOND DECIMAL PLACE**; e.g. XXXXX. XX).

31. The ratio of the 4th nearest distance of Cs^+ ion in CsCl structure to that of 5th nearest distance of Na^+ ion in NaCl structure is……
[Given: $\sqrt{2}=1.41, \sqrt{3}=1.73, \sqrt{5}=2.24, \sqrt{2.2}=1.48, \sqrt{11}=3.32$]

32. Given the following data,
For H_3PO_4; $pK_{a_1}=2.2$; $pK_{a_2}=7.2$; $pK_{a_3}=12.4$
And pK_{sp} of $Mg(NH_4)PO_4=12.6$ and the equilibrium concentration of $NH_4^+=0.1\,mol\ dm^{-3}$.
Calculate the value of $p(solubility)$ of $Mg(NH_4)PO_4$ at pH = 10; [Where 'p' means (- $\log_{10}$)]

33. The ratio of the number of triangular faces to that of the number of octagonal faces in a solid truncated cube is……(Round off your answer up to the second place of decimal)

34. If total number of peptide linkages in a decapeptide is 'x' and total number of $\alpha-$glucosidic linkages in a molecule of lactose is 'y', then the value of $\frac{(x)}{(y+2)}$ is…

35. $CH_2=C=CH_2 \xrightarrow{H_3O^+} \underset{(\text{Major product})}{(X)} \xrightarrow[\Delta]{\text{conc. } H_2SO_4} \underset{(\text{Major product})}{(Y)}$
Ratio of the total number of sigma bonds to that of pi bonds in a molecule of (Y) is……

36. Suppose your weight is 80 kg and you want to fly in the sky with the help of balloons each containing 50 moles of H_2 gas at 0.5 atm and 27°C. If 'x' is the minimum number of balloons that you need to attach with yourself to fly, then the value of $\frac{'x'}{4}$ is:
[Given: density of air = 1.25 g/L and it remains constant throughout. Also, neglect the weight of balloon]

Mathematics

PART – III

SECTION – A
(One or More than one correct type)

This section contains **06** multiple choice questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

37. If $a_1, a_2, a_3, \ldots\ldots a_n$ are in harmonic progression. Then which of the following is/are correct

(A) $\dfrac{(a_1a_3 + a_1a_2 - a_2a_3)(a_1a_2 + a_2a_3 - a_1a_3)}{(a_1\, a_2\, a_3)^2} = \dfrac{2}{a_2a_3} - \dfrac{1}{a_2^2}$

(B) $\dfrac{(a_1a_3 + a_1a_2 - a_2a_3)(a_1a_2 + a_2a_3 - a_1a_3)}{(a_1\, a_2\, a_3)^2} = \dfrac{(a_1 + a_3)(3a_1 - a_3)}{4a_1^2a_3^2}$

(C) Let $\beta_k = \dfrac{a_k}{\sum\limits_{k=1}^{n} a_k - a_k}$ for k = 1, 2, 3,....., n then $\beta_1, \beta_2, \beta_3, \ldots\ldots \beta_n$ are in harmonic progression

(D) Let $\beta_k = \dfrac{a_k}{\sum\limits_{k=1}^{n} a_k - a_k}$ for k = 1, 2, 3,....., n then $\dfrac{\beta_k\{\beta_{k-1} + \beta_{k+1}\}}{\beta_{k-1} \times \beta_{k+1}} = 2$ for k = 2, 3, ..., n – 1

38. Let n digit number is formed using digits from the set = {2, 3, 4} such that no two connective digits are 2. Let a_n represents such number which ends with 2, b_n represents such numbers which ends with 3 and c_n represents such numbers which ends with 4. Then which of the following is/are correct?

(A) $a_{n+1} = 2a_n + a_{n-1} \;\forall\; n \geq 2$

(B) $b_{n+1} = a_n + b_n + c_n \;\forall\; n \geq 1$

(C) $b_{n+1} = a_{n+1} + a_n \;\forall\; n \geq 1$

(D) $a_n = \dfrac{1}{2\sqrt{3}}\left\{\left(1+\sqrt{3}\right)^n - \left(1-\sqrt{3}\right)^n\right\}$

39. The sum of real part of non real roots of the equation $x^4 - 6x^3 + 10x^2 - 3x + a = 0$, $a \in R$, can be

(A) 6

(B) $\dfrac{3}{2}$

(C) 3

(D) $\dfrac{5}{2}$

40. Let $A = \left[a_{ij}\right]_{n\times n}$ be a $n \times n$ matrix with $a_{ij} = f(i)f(j) \;\forall\; 1 \leq i \leq n$ and $1 \leq j \leq n$, then which of the following is/are correct?

(A) If f(x) = x $\forall$ x $\in$ N then $\lim\limits_{n\to\infty} \dfrac{\left(\text{tr}\left(A^n\right)\right)^{\frac{1}{n}}}{n^3} = \dfrac{1}{3}$

(B) A^3 is symmetric matrix $\forall$ f(x) defined on natural numbers

(C) A^2 is singular matrix $\forall$ f(x) defined on natural numbers

(D) There exists function f(x) defined on natural numbers such that A^5 is non singular matrix

41. If ℓ_1, ℓ_2 and ℓ_3 are length of tangents drawn from I_1, I_2 and I_3 to the circumcircle of ΔABC (where I_1, I_2, I_3 are ex-centres of ΔABC opposite to A, B and C), then which of the following is/are correct?

(A) $\frac{\ell_1^2}{\ell_2^2} = \frac{r_1}{r_2}$

(B) $\ell_1\ell_2\ell_3 = abc\sqrt{\frac{R}{2r}}$

(C) $\frac{\ell_1^2}{\ell_2^2} = \frac{r_2}{r_3}$

(D) $\ell_1\ell_2\ell_3 = abc\sqrt{\frac{2r}{R}}$

42. Let $f : \{1, 2, 3, 4, 5, 6, 7\} \to \{1, 2, 3, 4, 5, 6, 7\}$, then which of the following is/are correct?

(A) The number of functions satisfying $f(f(f(x))) = x$ is 351

(B) Number of onto functions if $f(i) \neq i$ for $i = 1, 2, 3, 4, 5, 6, 7$ is
$7!\left\{\frac{1}{2!} - \frac{1}{3!} + \frac{1}{4!} - \frac{1}{5!} + \frac{1}{6!} - \frac{1}{7!}\right\}$

(C) The number of functions if $f(i) \geq f(j)\ \forall\ i > j$ is ${}^{13}C_7$

(D) The number of functions f if $f(i) \neq i$ for $i = 1, 2, 3, 4, 5, 6, 7$ is equal to
$7^7 - {}^7C_1 \cdot 6^6 + {}^7C_2 \cdot 5^5 - {}^7C_3 \cdot 4^4 + {}^7C_4 \cdot 3^3 - {}^7C_5 \cdot 2^2$

SECTION – B
(Single Digit Integer Type)

This section contains **06** questions. The answer to each question is a **Single Digit integer** ranging from **0** to **9**, both inclusive.

43. Let P be a point on $y = x$, with positive ordinate, such that angle APB is maximum, the radius of the circle passing through the points P, A and B is equal to r, then the value of $\sqrt{2}r$ is equal to (The coordinates of A and B are (1, 0) and (8, 0))

44. Total number of positive distinct real roots of the equation $\int_0^x \frac{t^{10}+1}{t^{10}+t^2+1}dt = x^2 + x$ is

45. Let $f : [0, 1] \to R$ be continuous function such that for any $x, y \in [0, 1]$ $xf(y) + yf(x) \leq 1$. If the maximum value of $\int_0^1 f(x)dx$ is k, then the value of $\frac{8k}{\pi}$ is

46. Let $f : [0, 1] \to R$ be differentiable function with continuous derivative $f'(x)$ on [0, 1] and $f(0) = 0$ and $f(1) = 1$, then the minimum value of $\left[3\int_0^1 \sqrt{1+x^2}\,(f'(x))^2\,dx\right]$ is (where [.] denotes greatest integer function)

47. Let $f:\left[1,3+2\sqrt{2}\right]\to R$ be a differentiable function, if f(1) = 0 and $f'(x)=\dfrac{1}{x^2+\dfrac{6}{\pi}\tan^{-1}\left(\dfrac{x+\dfrac{1}{x}}{2\sqrt{3}}\right)}$

$\forall\ x\in\left[1,3+2\sqrt{2}\right]$, then $\left[f\left(3+2\sqrt{2}\right)\right]$ is equal to (where [.] denotes greatest integer function)

48. The value of sin 4° sin 8° sin 16° sin 20° sin 28° sin 32° sin 40° sin 44° sin 52° sin 56° sin 64° sin 68° sin 76° sin 80° sin 88° is equal to N, then the value of $\left[2^{15}N\right]$ (where [.] denotes greatest integer function)

SECTION – C
(Numerical Answer Type)

This section contains **06 questions**. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **SECOND DECIMAL PLACE**; e.g. XXXXX. XX).

49. Let the sides of ΔABC are 3, 4, and 5 respectively and N be the centre of the circle passing through mid-point of AB, BC and CA, then the value of IN is equal to
(Where I is incentre of ΔABC)

50. In ΔABC, let BC = 2, CA = 4, AB = 5, then the value of
$a^3\cos 3B + 3a^2b\cos(A-2B) + 3ab^2\cos(2A-B) + b^3\cos 3A$ is equal to
(Where BC = a, CA = b, AB = c)

51. Let A, B and P lies on the ellipse such that AB is focal chord of the ellipse. The tangents drawn at A and B to the ellipse intersect at E(4, 2) and AP intersect corresponding directrix at F(4, –2). The line BP intersect the corresponding directrix at G(α, β), then the value of $|\alpha+\beta|$ is equal to
{Equation of the chord AB is y = x}

52. Let z_1, z_2, z_3 and z_4 be the roots of the equation $z^4 - 8iz^3 - 24z^2 + 32iz + 15 - i = 0$ and P(z) be a point on complex plane, then the minimum value of $\sum_{i=1}^{4}|z-z_i|$ is 2^x, then 8x is equal to

53. A fair coin is tossed repeatedly. If head is appeared, then one point is given to player A and O point to player B and if tail is appeared, then one point is given to player B and O point to player A. The game is won by the player who achieves a lead of two points first. The probability that A wins the game is

54. The value of $\lim\limits_{n\to\infty}\sum\limits_{k=1}^{n}\left(\sqrt{1+\dfrac{k}{n^2}+\dfrac{k^2}{n^3}}-1\right)$ is ℓ, then 12ℓ is equal to